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Ecological and functional diversity of gastropods associated with hermatypic corals of the Mexican tropical Pacific

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Abstract This study used multi-scale models to evaluate spatial variation in the structure of the assemblage and functional groups of gastropods associated with hermatypic corals of the Mexican Tropical Pacific. A geographic model examined the spatial scales of the region, geographic position and sites, while an oceanographic model considered the oceanographic characteristics and sites. The relationship between gastropods and environmental and spatial variables was analyzed with additive partitions, based on canonical correspondence analysis. Both models coincide in that the greatest variation of gastropods was presented at quadrant (i.e., small-scale) and site (i.e., local-scale) levels, and the lowest variation to the scale of geographic position and oceanographic characteristics (i.e., medium-scale). Abundance and Shannon diversity presented greater variation at small and local-scales, while species richness presented greater variation at medium-scale. The functional groups varied to the greatest extent at medium-scale in the geographic model and at local-scale in the oceanographic model. The highest abundance, richness and diversity were found at: (1) Plataforma Pavonas and Túnel Amarradero, ~20°N (Northern region), influenced by the California Current and the Mexican Coastal Current; (2) Caleta de Chon and Morro Potosí, ~17°N (Southern region), affected by the Mexican Warm Pool; and (3) Salchi and Isla Montosa, where relatively high biological productivity occur mainly due to coastal upwelling, ~15°N (Southern region). The carnivorous epifauna was the most important functional group at all studied scales. The most important species in both models were the corallivore *Coralliophila monodonta,* the symbiont *Muricopsis zeteki, Triphora hannai* of sandy substrate and *Vermicularia pellucida eburnea* and *Engina tabogaensis* of hard substrate. Variation among sites was explained mainly by the coverage of sessile taxa (e.g., percentage of coverage of the coral *Pocillopora* spp., bryozoans, hydrozoans, octocorals and sponges) and proportion of substrates (e.g., sand, rock, dead coral and rubble). However, at upwelling sites, other variables such as primary productivity and phosphate concentration also contributed to local-scale variation. At medium-scale, oceanographic characteristics explain the variation of the assemblage and functional groups of gastropods.

Keywords Marine gastropods · Coral ecosystems · Spatial variation · Northeastern

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Introduction

Coral ecosystems are characterized by providing refuge and zones for feeding, growth and reproduction for a large number of species. Mollusks are among the dominant invertebrates in these systems, which provide protection from physical disturbances and environmental stress, refuge from predation, reduced competition and increased local resources such as food and space (Kelaher et al. 2001; Stella et al. 2011). However, in all benthic communities, the physico-chemical variables and substrate types also influence the spatial distribution of the species (Neves et al. 2012).

The malacological assemblage depends on abiotic and biotic variables that operate at different spatial scales. Spatial ecology attempts to explain how the sum of small- and local-scale processes generates large-scale effects and vice versa (García 2006). For example, at large-scale (hundreds of kilometers), global variation in the climate and oceanic currents and their influence on the productivity and cycling of nutrients causes changes in the gastropod community structure that, in turn, influence their distribution and abundance (Chapman and Underwood 2008; Miloslavich et al. 2013). At medium- and local-scales (hundreds of meters to tens of kilometers), the availability of larvae and their recruitment and mortality are determinants in the composition and abundance of species. Local abundance of species changes hierarchically at these scales as a function of their size, mobility, life history and biological interactions (Underwood 2000). Finally, at small-scale (of the order of centimeters to meters), the behavior and structure of the habitat can control the spatial variation, although there is also evidence to suggest the importance of inter and intraspecific relationships (Chapman and Underwood 2008). For example, some small taxa with high rates of reproduction and micro-mollusks of hard substrates present high variability at small-scales (e.g., Morrisey et al. 1992; Olabarria and Chapman 2001; Cruz-Motta 2007; Chapman and Underwood 2008). However, species with low rates of growth, large colonies and stable populations present little spatial aggregation at small-scale (Smoothey and Chapman 2007; O'Leary and Potts 2011).

Spatial patterns that depend on the scale of distribution are important in all environments given their relationship to variation in biotic and abiotic conditions. This is important in the coastal zone, where the ecosystem undergoes frequent environmental change caused mainly by industrial, agricultural, fishery, mining and touristic activities that generate strong anthropogenic stress. For this reason, it is necessary to analyze the distribution patterns of the benthos in order to identify processes that determine the structure of the biological assemblages (Neves et al. 2012). In gastropods associated with hermatypic corals, it is expected that these trends will be maintained across a latitudinal gradient, i.e., at medium-scale, oceanographic characteristics and geographic position will explain differences in the assemblages. In contrast, at local- and small-scales, the physical and biological surroundings will have the greatest relationship with the assemblage and functional groups. This study conducted, for the first time, a multiscale analysis of the spatial variation of the ecological and functional diversity of gastropods associated with hermatypic corals in the region of the Mexican Tropical Pacific (MTP). Two multiscale models were constructed incorporating ecological and environmental factors that explain the variation of the gastropods. The geographic model analyzed spatial variation from the perspective of geographic-ecological division across a North-South latitudinal gradient. The oceanographic model considered a spatial analysis as a function of the influence of oceanographic characteristics on the distribution of organisms. The two models were complementary and provide information useful for decision-makers in the management and conservation of the coral ecosystems of the MTP, both at the local level for the creation and/or extension of Protected Natural Areas (PNA) and at the regional level for the proposal of PNA networks.

Material and methods

Study area

In this study, the MTP includes the coasts of Nayarit, Jalisco, Colima, Michoacán, Guerrero and Oaxaca, with an area of ~1500 km (Fig. 1). This region forms part of the Eastern Tropical Pacific (ETP). It is influenced oceanographically by the California Current (CC), the Mexican Coastal Current (MCC), the Cabo Corrientes Upwelling, the Mexican Warm Pool (MWP), the coastal upwelling of the Gulf of Tehuantepec (GT) and the Costa Rica Coastal Current (CRCC) (e.g., Wang and Enfield 2001; Kessler 2006; Fiedler and Talley 2006; Lavin et al. 2006; Godínez et al. 2010; Gómez-Valdivia et al. 2015; Reyes-Hernández et al. 2016). The CC is a cold current with sea surface temperatures (SST) of between 13 °C (winter) and 20 °C (summer). Within its zone of influence, masses of water of subarctic, subtropical and tropical origins converge (Fiedler and Talley 2006; Godínez et al. 2010). The MWP is part of the Western Hemisphere Warm Pool (WHWP, Wang and Enfield 2001), characterized by a SST above 28.5 °C and salinity of 34.5 ups. The wide annual variation in volume and temperature of the WHWP influences the rainy season, frequency and intensity of storms and hurricanes and plays an important role in the development of the El Niño-Southern Oscillation (ENSO) (Wang and Enfield 2001; Fiedler and Talley 2006). The waters of the Gulf of Tehuantepec are relatively productive due to the coastal upwelling and the turbulent mixing associated with the winds known as *Tehuanos* or *Nortes* that cross from the Gulf of Mexico to the Gulf of Tehuantepec (e.g., Stumpf and

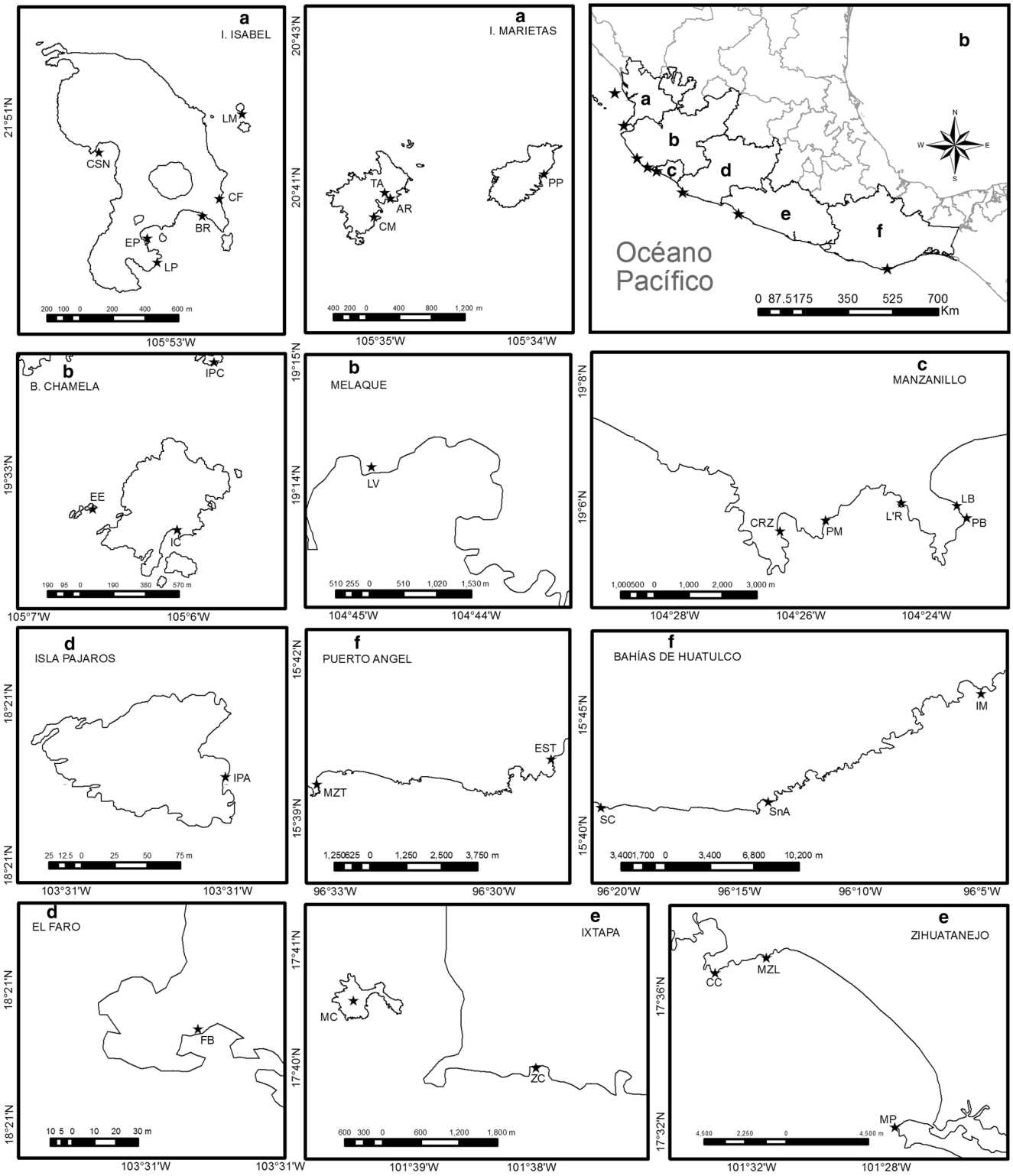

Fig. 1 Delimitation of the study area and location of the sampling sites in the PTM. **a** Nayarit: Isla Isabel, Caleta sin nombre: CSN, Las Monas: LM, Costa Fragata: CF, Bahía Rabijuncos: BR, Ensenada Pescadores: EP, Las Pozas: LP; Islas Marietas: Plataforma Pavonas: PP, Túnel Amarradero: TA, Área de Restauración: AR, Cueva del Muerto: CM; **b** Jalisco: Isla Pelícanos: IPE, El Elefante: EE, Isla Cocinas: IC, La Virgencita: LV; **c** Colima: Punto B: PB, La Boquita: LB, Paraíso Montemar: PM, L'Recif: L'R, Carrizales: CRZ; **d** Michoacán, Isla Pájaros: IPA, Faro de Bucerías: FB; **e** Guerrero: Morro Colorado: MC, Zacatoso: ZC, Manzanilla: MZL, Caleta de Chon: CC, Morro Potosí: MP; **f** Oaxaca: Isla Montosa: IM, San Agustín: SnA, Salchi: SC, Estacahuite: EST, Mazunte: MZT

Legeckis 1977; Trasviña et al. 1995; Barton et al. 2009). This is a prominent characteristic mainly from November to April–May.

The MTP is affected by El Niño (the warm phase of El Niño-Southern Oscillation or ENSO), which is characterized by a deep thermocline and nutricline that reduce primary productivity. These conditions are inverted with La Niña (the cold phase of ENSO) when deep water, rich in nutrients, rises to the surface favoring primary production (Fiedler and Talley 2006). In terms of the geomorphology, the MTP presents a narrow continental shelf, wide sandy and rocky beaches, zones of cliffs and protected bays that favor the discontinuous presence of coral ecosystems and the existence of a particular malacological fauna and a high number of endemisms (Reyes-Bonilla 1993; Glynn and Ault 2000)

This study explored 31 sites that present coral ecosystems with different degrees of reef development. These are found in Isla Isabel and Islas Marietas, Nayarit; Bahía Chamela and Melaque, Jalisco; Manzanillo, Colima; Faro de Bucerías and Caleta de Campos, Michoacán; Ixtapa and Zihuatanejo, Guerrero; Puerto Ángel and Bahías de Huatulco, Oaxaca (Fig. 1). These sites comprise rocky reefs and small fringing coral reefs with important coverages of coral, but with a low richness and equitivity of species due to the dominance of the genus *Pocillopora* (Glynn and Ault 2000; Reyes-Bonilla 2003; Reyes-Bonilla et al. 2010; López-Pérez et al. 2012; Reyes-Bonilla et al. 2013).

Field and laboratory work

In each site, four live coral samples of *Pocillopora* spp. were collected from randomly placed 0.25 m^2 quadrants, according to the criteria of Ramírez-Luna et al. (2002). The coral was fragmented and the live gastropods fixed with 70% ethylic alcohol and identified to the smallest taxon possible. Taxonomic identification of complete samples and those of size ≤0.5 mm were conducted following the criteria of Keen (1971) and Keen and Coan (1974). The scientific names were updated with WoRMS (2017), http://www.marinespecies.org/). The list of species was organized according to the criteria of Bouchet and Rocroi (2005) at the Family level. The collected specimens were incorporated into the mollusk collection of the Museo de Historia Natural de la Universidad del Mar (CMMHNUMAR) and that of the Laboratorio de Ecosistemas Marinos y Acuicultura (LEMA) of the Universidad de Guadalajara. In each quadrant, the total number of live individual species was recorded. The data were used to construct composition and abundance matrices. Whenever possible, each species was assigned to a functional group considering the association between life form (epifauna, semi-infauna, infauna) and feeding habit (carnivorous, herbivorous, scavenger or parasite). Where this information was not available for the species, the functional group was assigned at the Family level. In both cases, it was completed following the criteria of Keen (1971), Morton (1968), Vermeij (1978) and Levinton (2009), as well as using field observations made by the authors.

In each site, the structure of the benthic reef habitat was evaluated using five quadrants (each of 1 m^2) equidistantly located along 20 m transects, recording the coverage of live corals (*Pocillopora* spp., *Pavona* spp., *Porites* spp., *Psammocora* spp. and *Tubastraea* sp.), octocorals, bryozoans, hydrozoans, seaweed, sponges, sand, rock, dead coral and rubble. In each site, three to six transects were established depending on the size of the particular coral ecosystem.

Data analysis

The spatial variation of species richness (S), abundance (N), Shannon diversity (H′), assemblage of gastropods and functional groups was evaluated using two experimental designs. The geographic model (GM) considered the geographic-ecological division of the study area for conservation purposes. The factors considered were: (1) region (RE), with three levels; (2) geographic position (GP), with seven levels; and (3) sites (SI), with 31 levels. All the factors had random effects. The experimental design was a three-way unbalanced nested model, described with the expression $Y = \mu + RE_i + GP_j(RE_i) + SI_k(GP_j(RE_i)) + \varepsilon_{ijk}$. The factor RE_i represented the division of MTP in three regions, considering a geographic (Fouquet 2002) and ecoregions (Spalding et al. 2007) division: (1) North, which only considers the state of Nayarit that is part of the Cortezian ecoregion; (2) Central, which corresponds to the states of Jalisco, Colima and Michoacán that form the Mexican tropical Pacific ecoregion; and (3) South, which includes the states of Guerrero and Oaxaca that are within the Chiapas-Nicaragua ecoregion (Figs. 1 and 2). The factor GP_j corresponded to the geographic position located on a North-South latitudinal gradient, while the factor SI_k considered the 31 sampling sites (Fig. 1). Also, ε_{ijk} is the cumulative error of the model.

The oceanographic model (OM) considered the oceanographic characteristics described in the study area section. The CC has influence between 22°52′N and 18°45′N, on the coasts of Nayarit, Jalisco and Colima. In the OM the influence of the CC was considered in Nayarit. The MWP has an effect on the coasts of Jalisco, Colima, Michoacán, Guerrero and part of Oaxaca. The influence in the OM, was considered in Michoacán and Guerrero. The effects of these two oceanographic characteristics overlap among Jalisco, Colima, Michoacán and Guerrero, for which reason the influence of both (CC-MWP) was considered as an additional factor in Jalisco and Colima. The Gulf of Tehuantepec (GT) is influenced by different oceanographic conditions, such as Tehuanos winds and the Thermocline Bowl Tehuantepec (Fig. 2). The factors of the OM were an oceanographic

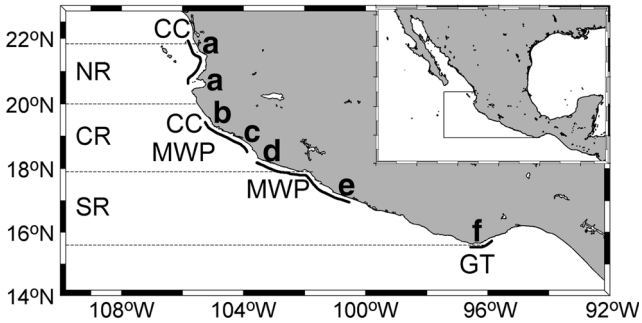

Fig. 2 Influence of spatial analysis models, geographic and oceanographic variation in the PTM. **a** Nayarit; **b** Jalisco; **c** Colima; **d** Michoacán, **e** Guerrero; **f** Oaxaca. Codes: **NR** North region, **CR** Central region, **SR** Southern region, **CC** California Current, **MWP** Mexican Warm Pool, **CC-MWP** California Current- Mexican Warm Pool, GT Gulf of Tehuantepec

characteristic (OC_i), with fixed effects and four levels, as well as sites (SI_j), with random effects and 31 levels. This was a two-way unbalanced nested model with mixed effects, described as: $Y = \mu + OC_i + SI_j(OC_i) + \varepsilon_{ij}$

Species richness (S), abundance (N) and Shannon diversity (H′, decits) were estimated per quadrant. For each of these, mean and standard error values were calculated per site, geographic position, oceanographic characteristic and region. The functional diversity of the gastropods was evaluated from the functional group assigned to each species. With the models described, the variation of the structural attributes (N, S and H ′), assemblage and functional groups of the gastropods were evaluated. Univariate (ANOVA) and multivariate (PERMANOVA) analyses of variance were used, based on permutations. The permutational ANOVA was used as a non-restricted method, due to the fact that the data did not fit parametric statistical assumptions. The ANOVA designs were constructed with Euclidean distance matrices (Anderson et al. 2008). The PERMANOVA were constructed with Bray-Curtis similarity matrices (Clarke and Warwick 2001). The significance of the ANOVA and PERMANOVA models was evaluated with a type III sum of squares (partial) and 10,000 permutations, under a reduced model (Anderson et al. 2008). *Post-hoc* analyses were only applied to the factor of the oceanographic characteristic (OC_i) of the Oceanographic Model (OM), due to their fixed effect. In order to estimate the contribution of the species that favored the dissimilarities among nested factors, a similarity percentage analysis (SIMPER) was used based on Bray-Curtis similarities (Clarke and Warwick 2001). SIMPER and PERMANOVA were performed with the same data pre-treatment where, prior to analysis, the data were fourth-root transformed to reduce the contribution of abundant species and to increase the contribution of the low abundance species. This strategy generates better estimates of the Bray-Curtis similarities (Clarke and Warwick 2001). These analyses were performed in the software Primer 6.1 + PERMANOVA (Anderson et al. 2008).

The relationship between the assemblage of gastropods and environmental-spatial variables was evaluated with a canonical correspondence analysis (CCA) at site, but considering the spatial variation throughout the study area (~1500 km). The CCA design was based on the results of the ANOVA and PERMANOVA, which evidenced the greatest significant differences in the assemblages of gastropods at the site or local-scale. For this, a unimodal relationship was assumed among the biological (Y), environmental (X) and spatial (W) variables. The biological variables corresponded to the composition and abundance of species and functional groups. The environmental variables were SST, primary productivity, phosphates, nitrates, dissolved oxygen and coverage of the benthic components. The SST, primary productivity, phosphates, nitrates and dissolved oxygen data were provided by Sergio F. Flores-Ramírez of the Center for Analysis and Conservation of Biodiversity of the Universidad Autónoma de Baja California Sur, while the coverage of the benthic components corresponded to our data collected during the sampling work. The spatial variables were obtained with a trend-surface analysis based on UTM geographic coordinates of latitude (x) and longitude (y) of each site, represented as a third-order polynomial ($x, y, x^2, y^2, x^2y, y^2x, x^3, y^3$) (Legendre and Legendre 2012).

In the CCA, a forward-selection procedure was used to identify which X and W variables best explained the variation of Y. The fit of the models was estimated as the proportion between the statistic Trace (i.e., the variation explained by the canonical axes) and the inertia of the model (i.e., the total variation of the biological matrix) (Legendre and Legendre 2012). Canonical additive partitioning was conducted to identify which spatial-environmental components explained the variation in the gastropods (Legendre and Legendre 2012). Colinearity between predictive variables was explored with Pearson correlations, eliminating the variables with $r \geq 0.90$. A variance inflation factor (VIF) ≤10 was also used. Statistical significance was tested with 10,000 permutations under a reduced model in the program CANOCO v4.5.

Results

Richness and abundance of species

A total of 1247 individuals were recorded, with a total richness of 72 species. The most abundant species were *Coralliophila monodonta* (Blainville, 1832) (N = 597), *Muricopsis zeteki* Hertlein & Strong, 1951 (N = 83), *Triphora hannai* F. Baker, 1926 (N = 53) and V*ermicularia pellucida eburnea* (Reeve, 1842) (N = 49). These species correspond to 64% of the total abundance of gastropods. The

species with the greatest frequency and distribution throughout the PTM reefs were *C. monodonta, M. zeteki, V. pellucida eburnea* and *Engina tabogaensis* Bartsch, 1931. Among the functional groups, the carnivorous epifauna presented the greatest abundance and richness of species in the factors of the study models (Table 1). The necrophagous infauna and herbivorous semi-infauna presented a low number of individuals and species. The highest diversity of functional groups was presented in the Northern region, Latitude 21° and in the oceanographic characteristic California Current. The Online Resource 1 presents the systematic list of living species of gastropods, collection sites and functional group assigned to each species.

Geographical model

The results evidenced significant differences in the abundance of gastropods among regions and sites. The components of variation had high values in the quadrants (i.e., small-scale) and sites (i.e., local-scale), while the lowest variation was presented in the geographic positions (i.e., medium-scale) (Table 2). In the Southern region, the highest abundances were found at Morro Potosí, Salchi, Estacahuite and Isla Montosa, while in the Northern region, the highest abundances were found at Tunel Amarradero (Fig. 1 and 3a). Mean species richness varied significantly with geographic position. The highest components of the variance were obtained in

Table 1 Species richness (S), abundance (N) and relative abundance (%) of the functional groups of the geographical and oceanographic models. NC: North region, CR: Central region, SC: South region, GP: geographic position, CC: California Current, MWP: the Mexican Warm Pool, GT: the coastal upwelling of the Gulf of Tehuantepec, CC-MWP: region overlapping the California Current and the Mexican Warm Pool, FG: functional group, Epi: epifauna, Semi: semi-infauna, Inf: infauna, Car: carnivore, Detr: detritivore, Filt: filtrador, Herb: herbivore, Par: parasite, Necr: necrophagous. Abbreviations of the models in Fig. 1and 2

Geographic model

FG	RE	NR			CR			SR			Total MTP		
		S	N	%	S	N	%	S	N	%	S	N	%
Epi	Car	12	172	57.5	18	76	43.4	27	628	81.2	**31**	**876**	70.2
	Detr	3	14	4.7	3	31	17.7	3	6	0.8	**3**	**51**	4.1
	Filt	4	18	6.0	6	37	21.1	8	58	7.5	**10**	**113**	9.1
	Herb	6	20	6.7	4	18	10.3	13	53	6.9	**19**	**91**	7.3
	Par	4	67	22.4	3	11	6.3	3	16	2.1	**6**	**94**	7.5
Semi	Herb	1	7	2.3	1	2	1.1	1	10	1.3	**1**	**19**	1.5
Inf	Necr	1	1	0.3				1	2	0.3	**2**	**3**	0.2
Total		**31**	**299**		**35**	**175**		**56**	**773**		**72**	**1247**	

FG	GP	21° 105°			20° 105°			19° 105°			19° 104°			18° 103°			17° 101°			15° 96°		
		S	N	%	S	N	%	S	N	%	S	N	%	S	N	%	S	N	%	S	N	%
Epi	Car	4	90	50.3	10	82	68.3	5	10	35.7	8	48	51.1	11	18	34.0	17	344	92.0	21	284	71.2
	Detr	2	12	6.7	1	2	1.7	2	5	17.9	2	21	22.3	1	5	9.4				3	6	1.5
	Filt	3	7	3.9	2	11	9.2	4	7	25.0	3	20	21.3	4	10	18.9	7	13	3.5	6	45	11.3
	Herb	4	14	7.8	4	6	5.0	2	5	21.4	1	6	5.3	2	7	13.2	6	13	3.5	11	40	10.0
	Par	3	51	28.5	2	16	13.3							3	11	20.8	1	2	0.5	3	14	3.5
Semi	Herb	1	4	2.2	1	3	2.5							1	2	3.8				1	10	2.5
Inf	Necr	1	1	0.6													1	2	0.5			
Total		**18**	**179**		**20**	**120**		**12**	**28**		**15**	**94**		**22**	**53**		**32**	**374**		**45**	**399**	

Oceanographic model

FG	OC	CC			CC-MWP			MWP			GT		
		S	N	%	S	N	%	S	N	%	S	N	%
Epi	Car	12	172	57.5	18	76	43.4	17	344	91.98	21	284	71.18
	Detr	3	14	4.7	3	31	17.7				3	6	1.50
	Filt	4	18	6.0	6	37	21.1	7	13	3.48	6	45	11.28
	Herb	6	20	6.7	4	18	6.3	6	13	3.48	11	40	10.03
	Par	4	67	22.4	3	11		1	2	0.53	3	14	3.51
Semi	Herb	1	7	2.3				1	2	0.53	1	10	2.51
Inf	Necr	1	1	0.3				1	2	0.53			
Total		**31**	**299**		**35**	**175**		**32**	**374**		**45**	**399**	

Table 2 Results of PERMANOVA designs, geographical and oceanographic models. N: Abundance, S: Species richness, H: Shannon diversity. RE: region, GP: geographical position, SI: sites, OC: oceanographic characteristics, Pse-F: Pseudo-F statistic, %CV: percentage of the components of variation. Significant differences (P ≤ 0.05) are in bold

	N			S			H′			Species assemblage			Functional groups		
Source of variation	Pse-F	*P*-value	%CV	Pse-F	*P*-value	%CV	Pse-F	*P*-value	%CV	Pse-F	*P*-value	%CV	Pse-F	*P*-value	%CV
Geographic model															
RE	13.299	**0.0186**	24.74	2.7682	0.1716	22.81	2.0456	0.2349	15.63	2.1	**0.0313**	16.41	1.505	0.2443	11.66
GP(RE)	0.2528	0.9014	0.00	2.9444	**0.0478**	20.86	1.5729	0.2129	13.80	1.9298	**0.0013**	16.24	1.8685	**0.0448**	16.73
SI(GP(RE))	3.2722	**0.0044**	34.45	1.0351	0.4302	5.35	2.0448	**0.0092**	25.69	1.4295	**0.0001**	18.08	1.2533	0.1148	15.74
Residuals			40.81			50.98			44.88			49.27			55.87
Oceanographic model															
OC	1.6631	0.1792	14.26	7.1446	**0.0019**	31.93	3.4158	**0.0399**	22.77	2.4154	**0.0005**	18.16	1.7233	0.0853	13.57
SI(OC)	3.2094	**0.0047**	38.89	1.0917	0.3675	9.84	1.9545	**0.0132**	27.26	1.6794	**0.0001**	25.79	1.4674	**0.0215**	23.88
Residuals			46.85			58.23			49.97			56.05			62.55

quadrants and regions, and the lowest in site-scale variation (Table 2). The highest values were estimated at 20°N (Islas Marietas) of the Northern region and 17°N (Caleta de Chon) and 15°N (Salchi) of the Southern region (Fig. 1 and 3b).

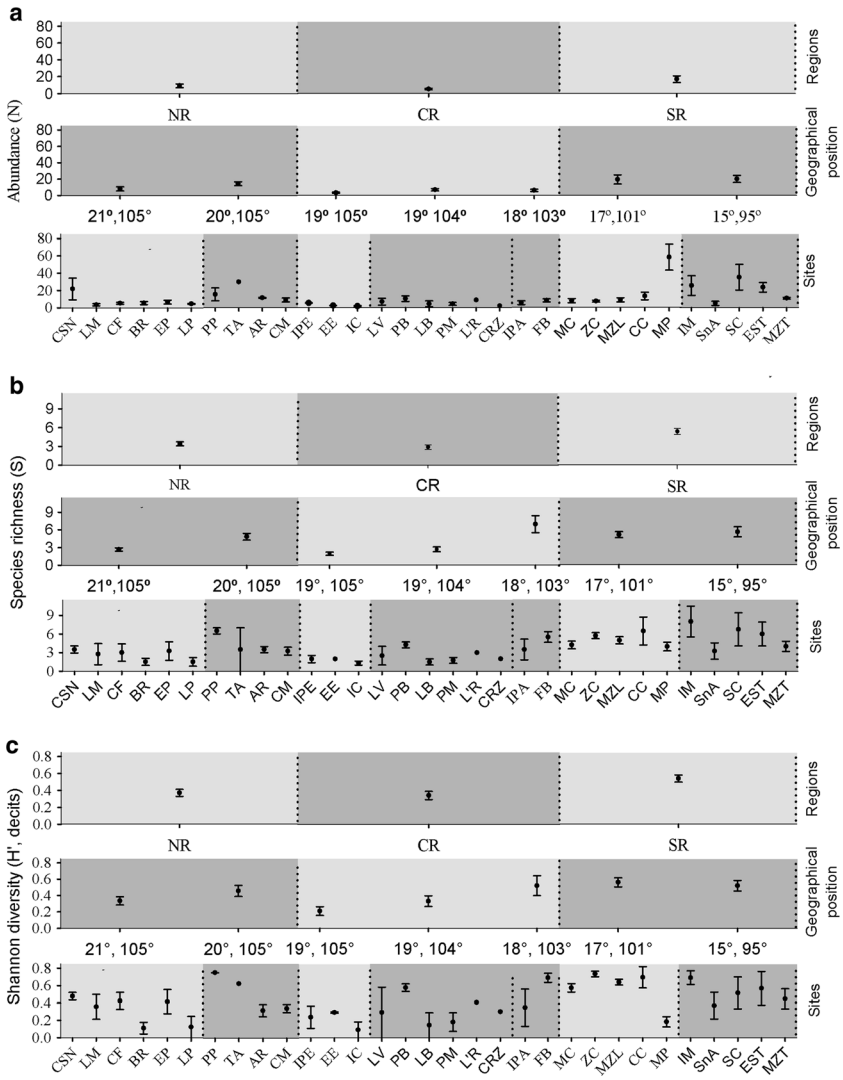

Fig. 3 Community attributes variation of coral reef-associated gastropod fauna, among geographic scales model. **a** Abundance (N); **b** Species richness (S); **c** Shannon diversity (H 'in decits). Site codes are given in Fig. 1

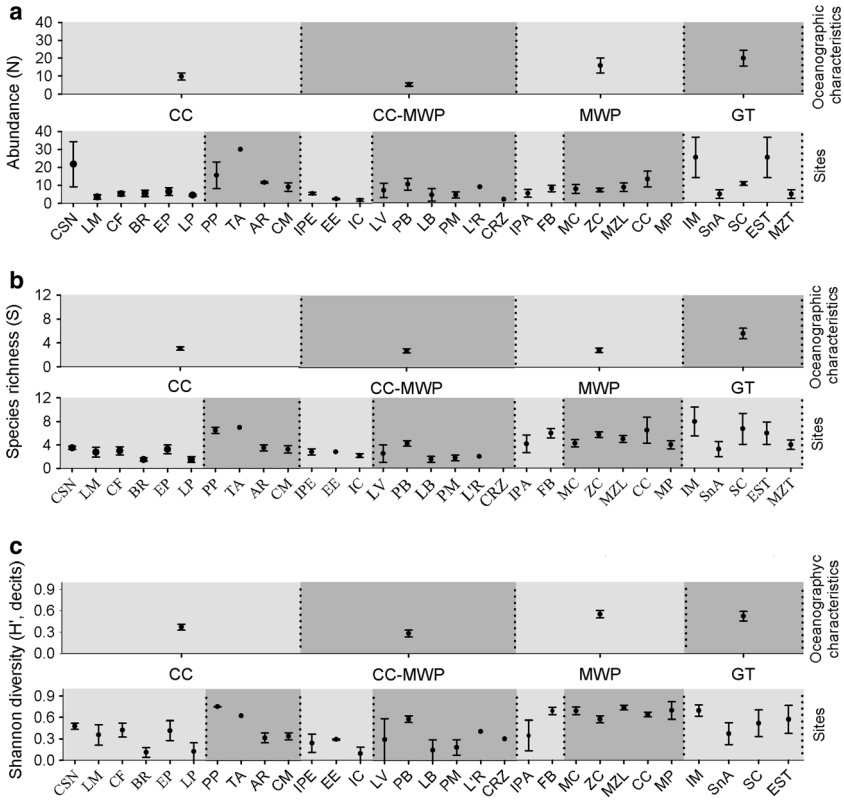

Fig. 4 Community attributes variation of coral reef-associated gastropod fauna, between the scales of the oceanographic model. **a** Abundance (N); **b** Species richness (S); **c** Shannon diversity (H 'in decits). Site codes are given in Fig. 1

Shannon diversity revealed significant differences at site level. The components of variation were larger between quadrants and sites, while geographic position presented the least variation (Table 2). The highest values were recorded in Plataforma Pavonas at 20°N, Zacatoso and Caleta de Chon at 17°N and Isla Montosa at 15°N (Fig. 1 and 3c). Comparison of the assemblage of species revealed significant spatial variation at the levels of region, geographic position and sites. The components of the variance had the same behavior as abundance and diversity; the high values were in quadrants and sites, while geographic position presented the lowest variation (Table 2). At the scale of region (i.e., large-scale), the species that favored dissimilarities among the assemblages were the corallivore *C. monodonta,* the symbionts *Gemophos sanguinolentus* (Duclos, 1833) and *M. zeteki* as well as the species characteristic of hard substrates *Hipponix antiquatus* (Linnaeus, 1767), *V. pellucida eburnea*, *Bostrycapulus aculeatus* (Gmelin, 1791) and *E. tabogaensis* and *Steironepion piperata* (E. A. Smith, 1882), a species of sandy substrates. The species that characterized dissimilarities among the scales of the model are presented in the Online Resource 2. The functional groups varied significantly at the scale of geographic position. As in species richness, high values of the components of the variance were presented in the quadrants and the geographic position, while the lowest variation was estimated at regional-scale (Table 2). In the Northern and Southern regions, the greatest richness of functional groups was recorded, with seven groups in each region. The carnivorous epifauna was the group of greatest richness and abundance across the spatial scales (Table 1, Online Resource 1). The herbivorous and filter feeding epifauna were abundant in the geographic positions of the Southern region, while the parasitic epifauna was important in the Northern region. In the Central region, few functional groups were identified, but the carnivorous, filter feeding and detritivorous epifauna were predominant (Table 1).

Oceanographic model

The oceanographic model showed that the abundance of the gastropods presented significant variation at level of sites. The highest variance components were obtained at the quadrant and site level (Table 2). Mean abundance varied among the sites with different oceanographic characteristics: Morro Potosí was influenced by the Mexican Warm Pool (MWP), Salchi located in the Gulf of Tehuantepec (GT) and Túnel Amarradero were influenced by the California Current (CC) (Fig. 2 and 4a). Species richness presented greater spatial variation at the level of oceanographic characteristics. The variance components were also high at this scale and in the quadrants (Table 2). The highest mean values of richness corresponded to Isla Montosa and Salchi, located in the GT (Fig. 2 and 4b). In the *post-hoc* analyses, differences in richness were maintained between CC and CC-MWP and GT and in the CC-MWP between

the MWP and GT (Online Resource 3). Shannon diversity presented significant differences in sites and oceanographic characteristics. The highest coefficients of variation were presented in the quadrants and sites (Table 2). The highest values of diversity corresponded to Plataforma Pavonas located in that influenced by the CC, Zacatoso and Caleta de Chon in the MWP and Isla Montosa in that influenced by GT (Fig. 2 and 4c). *Post-hoc* analyses of the Shannon diversity showed that the differences were maintained in CC-MWP with the MWP and the GT (Online Resource 3).

Analysis of the assemblage structure showed significant variation in the two spatial levels at oceanographic characteristics (i.e., medium-scale) and site-levels. These scales and the residuals had the highest coefficients of variation (Table 2). The *post-hoc* analyses showed an exception in the CC-MWP with the GT (Online Resource 3). The species that favored dissimilarities among the oceanographic characteristics were *M. zeteki, C. monodonta, B. aculeatus, S. piperata* and *V. pellucida eburnea*. In order of contribution, the species that marked the difference between sites were *C. monodonta, G. sanguinolentus, Pilosabia trigona* (Gmelin, 1791), *Cantharus elegans* (Griffith & Pidgeon, 1834), *V. pellucida eburnea* and *M. zeteki.* Functional groups varied at the level of sites. The coefficients of variation were, therefore, also high at site-scale and in the residuals (Table 2). Within these, the carnivorous epifauna was the group of greatest richness and abundance in the four oceanographic characteristics, the parasitic epifauna in the CC and the filter feeding epifauna in the CC-MWP, MWP and GT. Seven functional groups were identified as present in the CC and MWP (Table 1).

Canonical additive partition

The spatial variation of the assemblage and functional groups of gastropods at site level was explained mainly by environmental variables [*a*] (Table 3). In contrast, the variation explained jointly by the environmental and spatial variables [*b*], and the variation explained only by the spatial variation [*c*], made little contribution. Nevertheless, the residual variation of the models [*d*] reached values from 50 to 60%. In the species assemblage, the variation explained only by environmental variables [*a*] was 43% (Table 3). Primary productivity, coverage of *Pocillopora* spp., bryozoans, hydrozoans, sponge, sand, rock, dead coral, rubble and latitude (*x*) were the environmental and spatial variables that best explained variation in composition and abundance in the coral gastropods. In the functional groups, variation explained only by environmental variables [*a*] explained one third of the total variation of the gastropods (Table 3). This variation is mainly explained by the phosphates and the coverage values of octocorals, bryozoans, sponges and sand, as well as longitude (*y*).

Discussion

Models of spatial variation

The geographic and oceanographic models coincide in that the greatest variation of the assemblage of gastropods of the MTP occurs predominantly at quadrant or small-scale. But the most important spatial scales that support the ecological and functional diversity of gastropods are: (1) local-scale (e.g., among sites separated by tens of kilometers) and (2) medium-scale (e.g., among geographic positions and among oceanographic characteristics separated by hundreds of kilometers). In both models, abundance and diversity varied at local-scale and richness varied at medium-scale, while the functional groups varied at medium-scale in the geographic model and at local-scale in the oceanographical model. Accordingly, the greatest variation was presented at site and quadrant levels, while the lowest variation was obtained at medium-scale.

Mechanisms that generate large-scale variation in the organisms are the most accepted (Chapman and Underwood 2008) and best explained. However, it is difficult to identify the causes of variation in the gastropods at small-scale (1 to 10 m) (Olabarria and Chapman 2001). At local and medium-scale, important differences between sites and localities (10 to 35 km) have been reported for macrofauna of soft sediments (Morrisey et al., 1992), as well as in invertebrates of the rocky intertidal zone (Cruz-Motta 2007).

It is possible that, in the MTP, local-scale spatial variation of the gastropods of coral ecosystems is determined by the development and characteristics of the structure of the coral assemblage, which influences the heterogeneity of the habitat. Locally, this is defined by the spaces available among the branches and cavities of the coral, that increase the volume and space for larval development, refuge and food for juveniles and adults of many species, not only corallivorous symbionts, but also species of hard and even sandy substrates. In the Northern and Southern regions, true reefs are recognized as such because of their surface area, live coverage and density (Reyes-Bonilla 2003; López-Pérez et al. 2012). In the Central region, the sites present coral communities formed by small patches and/or rocky reefs with a structure and distribution that differs from those of the Northern and Southern regions (Reyes-Bonilla 2003). Possibly as a result of this, the species composition of the sites of the Central region corresponded to species of hard substrate, where *V. pelucida eburnea* characterized the coral substrate, with a contribution of 21.7%. This filter feeding species lives adhered to a hard substrate, or between gravel and small rocks. In contrast, the corallivorous species *C. monodonta* presented a low contribution (16%). Furthermore, the mean values of abundance, richness and diversity of coral gastropods were low among the sites and the geographic positions.

Table 3 Results of canonical additive partitioning based on CCA ordinations at level of species assemblage and functional groups of gastropods. The species assemblage and functional groups of gastropods correspond to matrices Y, Env. Var. = Environmental variables (matrix X), Spa. Var. = Spatial variables (matrix W), [*a*] = the variation of matrix Y explained by X alone, [*b*] = the variation explained jointly by X and W (i.e., the spatially-structured environmental variation), [*c*] = the variation explained by W alone, [*d*] = the residual variation, [*a* + *b* + *c*] = the total explained variation, P = *P*-value. Codes: P is concentration of Phosphates, PP is primary production; coverage benthos and substrates: Poc is *Pocillopora* corals, Bry is Bryozoa, Hyd is Hydrozoans, Sa is sand, Ro is rock, Spo is sponges, DC is dead coral, Rub is rubble, Oct is octocorales. The variation explained (VE) was estimated between the values of the Trace statistic and inertia of each model. Significant differences (P ≤ 0.05) are in bold. The environmental and spatial variables shown in italics represent that they alone had a significant statistical contribution within CCA models

	[*a*]	[*b*]	[*c*]	[*d*]	Total [*a* + *b* + c]	Env. Var. [*a* + *b*]	Spa. Var. [*b* + c]
Species assemblage	42.7	6.3	0.3	50.7	VE = 0.4927	VE = 0.4901	VE = 0.0658
					P = **0.0008**	*P* = **0.0017**	*P* = **0.0030**
					Env. Var.: PP, *Poc,* Bry, Hyd, Spo, Sa, Ro, DC, Rub		
					Spa. Var.: Latitude (*x*)		
Funcional groups	26.6	10.5	2.1	60.7	VE = 0.3921	VE = 0.3711	VE = 0.1263
					P = **0.0026**	*P* = **0.0069**	*P* = **0.0303**
					Env. Var.: P, Oct, Bry, Spo, Sa		
					Spa. Var.: Longitude (*y*)		

The oceanographic model at medium-scale showed that the dominant species of the coral assemblage were the corallivorous *C. monodonta* and the symbiont species *G. sanguinolentus, A. elegans* and *M. zeteki* that, due to their dietary and habitat specificity, are linked to the coral reef in a close predator-prey relationship. These species are of wide distribution in the reefs of the ETP (Guzmán 1988). For this reason, their abundance in the MTP is not unexpected. The species *C. monodonta* was among the most abundant in the MTP and characterized the reefs of the Northern and Southern regions, with a contribution ranging from 50 to 80%.

In general, variation of the assemblage and richness of gastropods at medium-scale in the oceanographic model could be controlled by external factors such as the environmental conditions, availability of food and hydrodynamics that facilitate the passive local dispersion of larvae. This is important both in terms of the duration of the gastropod larval stages in the water column and the availability of substrate for the adult. These are determinant factors in the species assemblage structure (Kempf 1981; Campos and Díaz 2007). Variation at medium-scale can also be explained by the physical properties of the seawater that condition other biological processes of the organisms. For example, SST influences the physiology and affects distribution and abundance at different spatial scales (Miloslavich et al. 2013). The oceanographic characteristics of the MTP are affected by other oceanographic conditions that influence the ETP, such as the warm phase of the ENSO (Wang and Enfield 2001). In particular, coral bleaching has been reported during El Niño events, (e.g., López-Pérez et al. 2016), which could affect the composition and abundance of mollusks in the MTP (Miloslavich et al. 2013). In some sites of the study area, different degrees of damage associated with the ENSO were observed in the reefs and coral communities (Reyes-Bonilla et al. 2003). The structure of the gastropod assemblage is, therefore, affected by the reduction of live coral cover and the composition of coral species, causing variation in the assemblage of gastropods at the scales of sites and oceanographic characteristics.

This study also showed that there are important variations in the assemblage and abundance of gastropods at regional-scale. However, richness, diversity and functional groups did not vary significantly. The gastropods, therefore, present a similar pattern of distribution and species composition at large-scale within the MTP, since the reefs and coral communities of this region present a semi-continuous distribution across a latitudinal gradient, which suggests high connectivity among the study sites and a possible shared evolutionary history. Regardless of the differences in the composition of species among regions, the structure of the assemblage can be similar. The elements that favor variation at greater scales are depth, temperature, water movement and sediment characteristics, as well as the level of human impact (Cruz-Motta 2007; Kraufvelin et al. 2011; Neves et al. 2012).

The functional groups varied significantly at the scales of the geographic position in the geographic model and of sites in the oceanographic model. In both cases, the components of variation were high among quadrants, followed by sites in the OM and geographic position in the GM. Also in both models, carnivorous epifauna formed the dominant functional group. In the coral substrate, this group is common due to the presence of corallivorous gastropods, symbionts and species that feed on carnivorous and herbivorous microgastropods and other organisms. This study also evidenced that the functional groups of detritivorous and herbivorous epifauna presented a low richness and abundance, mainly in the Central region. In contrast, in the zone of influence of the CC-MWP, the values

of the detritivorous epifauna were high. Despite the fact that the sampling technique used and the substrate studied did not comprise all of the functional groups of gastropods, the diversity of life forms in the coral was high, which is explained by the high values of abundance, richness and diversity (Neves et al. 2012). This coincides at the scale of site, geographic position, oceanographic characteristic and region, where the greatest richness, abundance and diversity were estimated. It is possible that this is related to the upwellings, which have a significant effect on the supply of nutrients towards the euphotic zone, concentrations of chlorophyll and the rates of primary productivity that perhaps favor higher biodiversity. In the MTP, there are two zones of upwellings in Cabo Corrientes, which coincide with the area of influence of the CC and the Gulf of Tehuantepec (Trasviña et al. 1999; Kessler 2006; Lavin et al. 2006; Zamudio et al. 2007; Barton et al. 2009). The highest diversity of gastropod functional groups was identified in these areas.

Canonical additive partition

This analysis showed that the assemblage and functional groups of gastropods were explained only by the environmental variation [*a*] at the site level. This indicates that heterogeneity in habitat composition (benthic groups) is important in terms of the spatial variation of the gastropods. The variation explained simultaneously by the biological and spatial variables [*b*] and that explained only by the spatial variables [*c*] had low values, possibly due to the ecological relationships with other biotic and abiotic processes that occur in the reef community or in the adjacent ecosystems that were not considered in this study. The high values of the residual variation [*d*] suggest that the lack of recording of other variables could have influenced the variation of the gastropods at local-scale.

The environmental variables that explained the greatest variation of the assemblage of gastropods were: coverage by sessile taxa (*Pocillopora* spp., bryozoans, hydrozoans, sponges), proportion of substrates (sand, rock, dead coral, rubble) and primary productivity. This evidences that the factors that determine spatial variation are related to the physical and biological environment of the communities and to significant environmental variables in the distribution and biology of the studied group. In hard substrates such as coral, spatial variation could depend on the different coverage by sessile taxa, the proportion and physical and chemical characteristics of the substrate, primary productivity, dissolved oxygen, phosphates and nitrates that favor a microbial coverage that provides food to assemblages of meiofauna (Olabarria et al. 1998; Cruz-Motta 2007; Neves et al. 2012). The composition of the microbial assemblages associated with the mucus of coral is associated with the coral species mainly as a function of time. It is, therefore, possible that this association may be determined by environmental factors such as the oceanographic characteristics, structure of the benthic community and the SST (Ceh et al. 2011). These changes in the microbial assemblages can indirectly influence the composition and abundance of some species of gastropods.

In conclusion, the multiscale models that analyzed the change in the ecological diversity of the gastropods associated with coral ecosystems demonstrated patterns of spatial variation at the local-scale driven by the habitat structure, represented by the coverage of the coral *Pocillopora* spp., bryozoans, hydrozoans, octocorals and sponges, as well as the proportion of substrates, such as sand, rock, dead coral and rubble, present in the reef, as well as the oceanographic variables of primary productivity and phosphates. It is recommended that future studies consider environmental conditions at the microhabitat level for both biotic and abiotic components. This may permit a better understanding of the assemblage of gastropods associated with coral ecosystems, given the importance of variability at local-scales, which could be confused with the variation that occurs at larger (i.e., medium and large) scales. This aspect could be of utility for decision-making in management concerning coral communities, and could contribute to a better interpretation of surveillance programs that include the detection of anthropogenic impacts and the incorporation of spatial scales into questions of biodiversity and marine conservation.

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Compliance with ethical standards

Conflict interest The authors declare that they have no conflict of interest.

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